

TEAM ASTRA • SIH INTERVIEW MASTER PREP

ANNAPURNA — Smart India Hackathon

Master Q&A Guide

Comprehensive, Code-Grounded Defenses for 57 SIH Judge Questions • ITER Ladies
Hostels LH1–LH5, Bhubaneswar

Project Tech Stack & Architecture Overview

- **Frontend:** React 18 + TypeScript + Vite + Tailwind CSS
- **UI & 3D Graphics:** Three.js (3D Holographic AI Emblem Canvas), Matter.js (Bubblegum Physics Canvas), Recharts, Lucide Icons
- **AI & ML:** Multivariate Linear Regression ML Engine (MLSurplusPredictor.ts, $R^2=94.2\%$) + Google AI Studio (Gemini 1.5 Flash API endpoint)
- **Accessibility:** Web Speech API (VoiceAssistWidget.tsx text-to-speech)
- **Data Schema:** PostgreSQL Normalized Schema (supabase_schema.sql) with 8 core tables + LocalStorage fallback store (store.ts)

1. Problem Statement & Idea Validation

Q1: What exact problem statement (PS number/theme) does this map to, and how does your solution address every clause of it?

ANNAPURNA maps to the **Smart Campus & Food Sustainability** theme.

- *Clause 1 (Over-preparation & Guesswork):* Solved via MLSurplusPredictor.ts using multivariate linear regression on turnout & temperature history to calculate prep sizes with a 2.5% safety buffer.
- *Clause 2 (Real-time Turnout Visibility):* Solved via dynamic QR turnstile check-ins and live attendance meters.

- *Clause 3 (Unhandled Surplus Food)*: Solved via our 4-step Surplus Redistribution module connecting mess staff to Robin Hood Army & Feeding India.

Q2: How did you validate that food wastage/demand-mismatch is a real, measured problem on your campus?

In ITER Ladies Hostels (LH1 to LH5, housing 2,500+ female students), mess contractors currently prepare food for 100% capacity regardless of day of week. On weekends and before holidays, student turnout drops by 25%–40%, resulting in **35 kg to 60 kg of unserved food wasted per mess daily**. We validated this through interviews with hostel wardens, mess staff, and 120+ resident students.

Q3: Why QR check-in specifically over biometrics, RFID, or geofencing?

- *Biometrics (Fingerprint/Face)*: Slow (>3s/person), hygienic risks with wet/messy hands, expensive hardware.
- *RFID Cards*: High card issuance costs, frequent loss by students, card-sharing proxy risks.
- *Geofencing*: GPS inaccurate inside multi-story RCC hostel buildings.
- *Dynamic QR*: Zero hardware cost, sub-second scan speed, and rotating TOTP session tokens that prevent screenshot proxy sharing.

Q4: Who are your actual end users?

1. **Students (LH1–LH5 Residents)**: Need fast (<1s) check-in, transparent daily menus, and feedback reviews.
2. **Mess Contractors & Chief Wardens**: Need accurate prep forecasts 3 hours prior, waste logs, and cost analytics.
3. **NGO Food Rescue Partners**: Need real-time alerts when verified unserved surplus food is packaged for pickup.

Q5: What's the single biggest inefficiency this removes that manual registers can't?

ANNAPURNA acts **before cooking begins**. Manual registers only log attendance after food is already prepared and wasted. Our ML Demand Engine tells mess staff 3 hours

prior to meal service exactly how many students will attend based on historical patterns and weather, preventing over-preparation at the source.

Q6: Is this college-specific or generalizable?

100% generalizable to any institutional dining facility (universities, corporate canteens, military messes). Our database schema contains multi-tenant messes and `institution_id` structure.

2. Technical Architecture & Database

Q7: Walk through the end-to-end flow from QR scan to dashboard update.

1. Authority screen generates dynamic QR token (`qrcode.react`).
2. Student app scans token via camera or tap-checkin.
3. Client validates token expiration and checks unique constraint (`student_id`, `meal_id`, `date`).
4. `appStore.checkInStudent()` mutates state, saves record, and notifies UI subscribers.
5. Live Attendance & Authority meters update in real-time.

Q8: Why Supabase over custom Express + Postgres or Firebase?

Supabase provides auto-generated REST/GraphQL APIs via PostgREST, native PostgreSQL Row-Level Security (RLS) policies for student/authority data isolation, and real-time WebSocket subscriptions without writing boilerplate backend controllers.

Q9: How do you prevent screenshot QR sharing for proxy check-in?

The authority QR code rotates its session token hash every 15–30 seconds. Screenshots taken by students expire almost immediately and fail validation.

Q10: How do you prevent duplicate check-ins for the same meal?

Enforced at database and state level via composite unique key `UNIQUE(student_id, meal_id, checkin_date)`. Second scan attempts return an immediate validation error.

Q16: How is your forecasting calculated?

We implemented a custom **Multivariate Linear Regression Engine (MLSurplusPredictor.ts)** trained on day-of-week, ambient temperature, and turnout history, returning $R^2=94.2\%$ accuracy score and recommended prep quantities with a 2.5% safety buffer.

3. Data Science & Machine Learning

Q21: Is there an actual ML model in this system? Be precise.

Yes. `MLSurplusPredictor.ts` implements a multivariate linear regression algorithm computing feature weights, Mean Squared Error (MSE), and R^2 accuracy score. It includes an interactive dataset trainer accepting CSV/JSON online uploads, plus integration with **Google AI Studio (Gemini 1.5 Flash API)** for natural-language zero-waste recipe recommendations.

Q23: What do you do during the cold-start period before data accumulates?

During the first 14 days, the system applies a heuristic fallback: 85% capacity for weekdays and 70% for weekends, transitioning automatically to full ML regression once 30+ meal records accumulate.

4. Feasibility, Business Model & Impact

Q38: What is your monetization / sustainability plan?

B2B Institutional SaaS Subscription: **₹15–₹20 per student per month** (approx. ₹15,000/year per hostel block) paid by the university administration or mess catering contractor.

Q43: What is the ROI pitch to a mess contractor?

A 500-student hostel mess wastes ~40 kg food daily (₹2,000/day = ₹60,000/month). ANNAPURNA reduces over-preparation by 60%, saving **₹36,000/month** against a ₹3,000 monthly SaaS fee — delivering a **12x monthly ROI**.

Q47: What is the projected environmental impact?

Preventing 1 kg of food waste avoids 2.5 kg CO₂e emissions and saves 1,000 L of embedded water. Scaling across LH1–LH5 saves **~15 tonnes of food waste annually**, preventing **37.5 tonnes of CO₂ emissions**.